

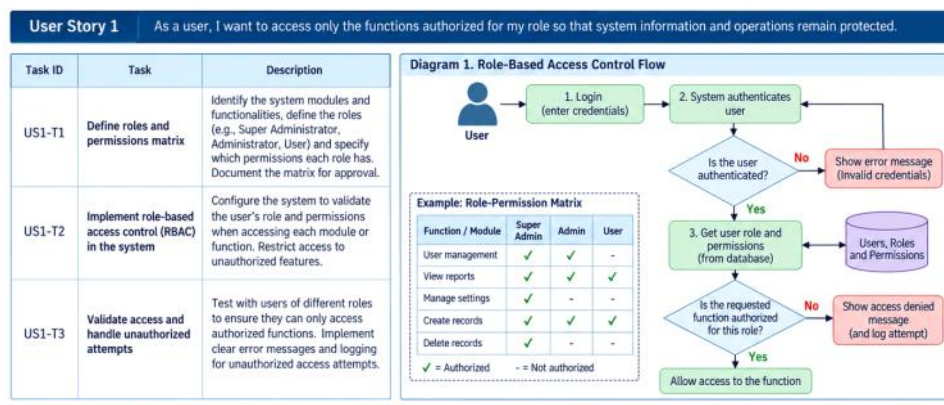
ID	Task	Description
US1-T1	Define roles and permissions matrix	Identify the different system roles, such as Super Administrator, Administrator, and User. Define which modules and functions each role is allowed to view, create, modify, or delete.
US1-T2	Design role-based access control (RBAC)	Design the process that identifies the user's role after login, retrieves the corresponding permissions, and determines which modules and functions the user can access.
US1-T3	Validate authorized and unauthorized access	Design test cases to verify that authorized users can access their permitted functions while unauthorized users receive an Access Denied message. Direct access attempts through URLs should also be tested.

Proposed process:

User Login → Authentication → Get User Role → Retrieve Permissions → Request Function → Validate Permission → Allow Access / Access Denied

Example permission matrix:

Function	Super Admin	Admin	User
User Management	✓	✓	X
View Reports	✓	✓	✓
Manage Settings	✓	X	X
Create Records	✓	✓	✓
Delete Records	✓	X	X



ID	Task	Description
US2-T1	Design the administrator management interface	Design the screens required to display the administrator list and provide options to Create, Edit, Activate, and Deactivate administrators. Define fields such as name, email, role, and status.
US2-T2	Design administrator CRUD logic and validations	Design the logic required to create, read, update, activate, and deactivate administrator accounts. Include validations for required fields, duplicate emails, valid roles, and database storage.
US2-T3	Validate	Design test cases for creating, modifying, activating, and

functionality and restrict access

deactivating administrators. Verify that only users with the **Super Administrator** role can access the Administrator Management module.

Proposed process:

Super Administrator → Administrator Management → Display Administrator List → Select Action

From there, the process can follow three main paths:

Create Administrator:

Create → Enter Administrator Information → Validate Data → Save to Database → Display Success Message

Modify Administrator:

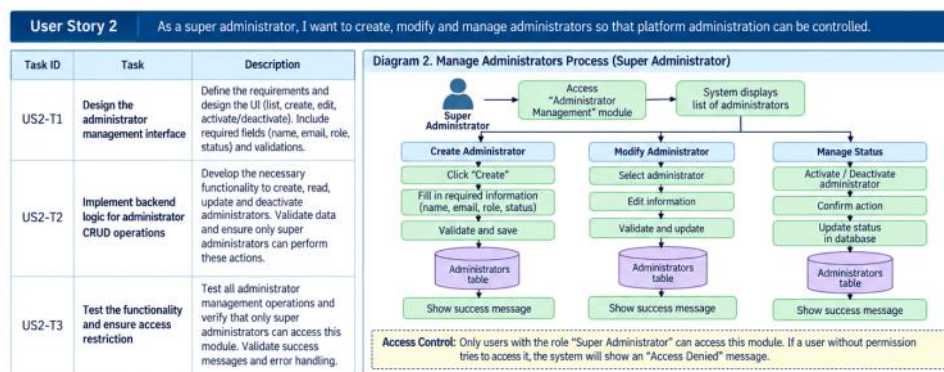
Select Administrator → Edit Information → Validate Changes → Update Database → Display Success Message

Manage Administrator Status:

Select Administrator → Activate/Deactivate → Confirm Action → Update Database → Display Success Message

Before accessing the module, the system should also perform:

Check User Role → Is Super Administrator? → Yes: Allow Access / No: Access Denied



Test Case

Flow

Expected Result

TC-US2-01 – Create administrator successfully

Super Admin logs in → Opens Administrator Management → Clicks **Create Administrator** → Enters valid name, email, role, status and password → Clicks **Save Administrator**

The system validates the information, creates the administrator and displays a success message.

TC-US2-02 – Create administrator with missing fields

Super Admin → Create Administrator → Leaves a required field empty → Clicks **Save Administrator**

The system does not create the administrator and indicates which required fields must be completed.

TC-US2-03 – Create administrator with duplicate email

Super Admin → Create Administrator → Enters an email already registered → Clicks **Save Administrator**

The system rejects the operation and displays a message such as **"Email already registered."**

TC-US2-04 – Edit administrator successfully

Super Admin → Administrator List → Selects **Edit** → Modifies information → Clicks **Update Administrator**

The information is validated and updated in the database. A success message is displayed.

TC-US2-05 – Edit administrator with invalid data

Super Admin → Administrator List → Edit → Enters an invalid email or removes a required field → Clicks **Update Administrator**

The system rejects the update and displays the corresponding validation message.

TC-US2-06 –

Super Admin → Administrator List → Clicks

The administrator status

Deactivate administrator	Deactivate → System asks for confirmation → Confirms action	changes from Active to Inactive and the system displays a success message.
TC-US2-07 – Cancel administrator deactivation	Super Admin → Administrator List → Clicks Deactivate → Confirmation window appears → Clicks Cancel	No changes are made to the administrator account.
TC-US2-08 – Activate administrator	Super Admin → Administrator List → Selects an inactive administrator → Clicks Activate → Confirms action	The administrator status changes to Active .
TC-US2-09 – Unauthorized user tries to access module	Administrator/User logs in → Attempts to access Administrator Management	The system detects that the user is not a Super Administrator and displays Access Denied .
TC-US2-10 – Unauthorized direct URL access	Administrator/User manually enters the Administrator Management URL	The backend validates the user's role before displaying the page and blocks access with 403 / Access Denied .
TC-US2-11 – Administrator list loads correctly	Super Admin → Opens Administrator Management	The system displays registered administrators with Name, Email, Role, Status and available actions.
TC-US2-12 – Search administrator	Super Admin → Administrator List → Enters a name or email in the search field	The system displays only administrators matching the search criteria.

Test Flow 1 — Create Administrator

Start



Super Administrator logs in



Access Administrator Management



Click **Create Administrator**



Enter administrator information



Are all required fields valid?

No → Display validation errors → Return to form

Yes → Check email availability



Is email already registered?

Yes → Display duplicate email error

No → Save administrator in database



Display success message



End

Test Flow 2 — Modify Administrator

Start



Super Administrator logs in

↓
Access Administrator Management
↓
Select administrator
↓
Click **Edit**
↓
Modify information
↓
Validate information
↓
Is the information valid?
No → Display validation error → Return to form
Yes → Update database
↓
Display success message
↓
End

Test Flow 3 — Activate / Deactivate Administrator

Start
↓
Open Administrator Management
↓
Select administrator
↓
Click **Activate / Deactivate**
↓
Display confirmation dialog
↓
Does the Super Administrator confirm?
No → Cancel operation → Return to administrator list
Yes → Update administrator status
↓
Save change in database
↓
Display success message
↓
End

Test Flow 4 — Access Control

Start
↓
User logs in
↓
User requests Administrator Management
↓
System retrieves user's role
↓
Is role = Super Administrator?
No → Block access → Display **Access Denied / 403** → End
Yes → Display Administrator Management module
↓
End

As a user, I want to access only the functions authorized for my role so that system information and operations remain protected.

Test Case	Flow	Expected Result
TC-US1-01 – Valid login	User enters valid credentials → System authenticates user	The system allows login and retrieves the user's assigned role.
TC-US1-02 – Invalid login	User enters incorrect credentials → Attempts to log in	The system rejects the login and displays Invalid credentials .
TC-US1-03 – Super Administrator accesses authorized function	Super Administrator logs in → Opens an authorized module	The requested module is displayed correctly.
TC-US1-04 – Administrator accesses authorized function	Administrator logs in → Opens a module authorized for the Administrator role	Access is granted.
TC-US1-05 – Regular user accesses authorized function	User logs in → Opens a permitted module	Access is granted and the function can be used normally.
TC-US1-06 – User attempts unauthorized module access	User logs in → Attempts to open a module not included in their permissions	The system blocks access and displays Access Denied .
TC-US1-07 – Unauthorized direct URL access	User manually enters the URL of a restricted module	The backend checks permissions and returns 403 / Access Denied .
TC-US1-08 – Unauthorized function through UI	User opens an authorized module → Attempts to execute a restricted action such as Delete	The system blocks the action even if the user can access the module.
TC-US1-09 – Restricted menu options are hidden	User logs in → System loads the main menu	Functions that are not authorized for the user's role are hidden or disabled.
TC-US1-10 – Role permissions are updated	Super Administrator modifies the permissions assigned to a role → User logs in again	The user's available functions reflect the updated permission configuration.
TC-US1-11 – User has no assigned role	User logs in successfully but does not have a valid role assigned	The system blocks access to protected functions and reports a configuration/access error.
TC-US1-12 – Session permissions validation	User logs in → Session is created → Attempts several operations during the session	Every protected operation checks the user's permissions before execution.

Test Flow 1 — Authentication and Role Retrieval

Start

↓

User enters credentials

↓

System authenticates credentials

↓

Are credentials valid?

No → Display **Invalid Credentials** → End

Yes → Retrieve user information

↓

Retrieve assigned role

↓

Retrieve permissions associated with role

↓
Create authenticated session
↓
Display authorized interface
↓
End

Test Flow 2 — Authorized Function Access

Start
↓
Authenticated user selects a function
↓
System retrieves current user role
↓
System checks role permissions
↓
Is the requested function authorized?
No → Block access → Display **Access Denied** → End
Yes → Allow access to requested function
↓
Execute function
↓
End

Test Flow 3 — Direct URL Unauthorized Access

Este es importante porque no basta con ocultar botones.

Start
↓
User is authenticated
↓
User manually enters restricted URL
↓
Server receives request
↓
Retrieve authenticated user's role
↓
Check permission for requested route
↓
Does the user have permission?
Yes → Load requested page
No → Reject request
↓
Return **403 Forbidden / Access Denied**
↓
Optionally record unauthorized access attempt
↓
End

Test Flow 4 — Restricted Action Inside an Authorized Module

Por ejemplo, que un usuario pueda ver registros pero no eliminarlos.

Start
↓
User accesses authorized module
↓
User selects an action such as **Delete Record**
↓

System checks permission for that specific action

↓

Is Delete permission assigned?

No → Reject operation → Display **You do not have permission to perform this action**

Yes → Execute deletion

↓

Update database

↓

Display success message

↓

End

Test Flow 5 — Menu Visibility According to Role

Start

↓

User logs in

↓

System obtains user's role and permissions

↓

Load application menu

↓

For each module/function

↓

Is function authorized?

Yes → Display menu option

No → Hide or disable menu option

↓

Display final authorized menu

↓

End

Test Flow 6 — Permission Change

Start

↓

Super Administrator modifies role permissions

↓

Save new permissions

↓

User starts a new session

↓

System retrieves updated permissions

↓

User requests a function

↓

Is the function still authorized?

Yes → Allow access

No → Block access

↓

Display **Access Denied**

↓

End